Sāgarasaṅgama: Fathoming the Sanskrit E-Text Oceans

Tyler Neill
Brooklyn, NY
tyler.g.neill@gmail.com

Abstract

Sanskrit e-texts are now used at a scale they never were before, by aggregated corpora, full-text search engines, and language models, yet no one knows how many exist. Five very large online collections alone hold some 22,000 texts and 4.9 GB of IAST plaintext, many times the last published estimate. The aggregated collections built by search and NLP projects are not good representations of the material available. This paper presents Sāgarasaṅgama, a set of Atlases that map and measure these five collections without hosting any of their text, and reports what they reveal about the collections' size, growth, and hidden content. It then sketches how Sāgarasaṅgama will merge with the Sanskrit E-Text Inventory (SETI), which registers e-texts and links them to work records in the Pandit database, to give a single metadata-based inventory of all extant Sanskrit e-texts — a useful denominator against which any collection or meta-collection can be measured. Finally, it looks ahead to a second denominator: all premodern works that could one day have an e-text, to be obtained by merging the expanded SETI onto the New Catalogus Catalogorum.

1 Introduction: the state of Sanskrit e-text

A great deal of Sanskrit literature with printed editions has already been digitized as searchable text. Three large high-quality collections, E-bharatisampat, Sanskrit Wikisource, and GRETIL, together hold about 10,000 texts in 3.4 GB of IAST plaintext (§2). These collections were built independently over three decades by universities, foundations, volunteer communities, and individual scholars, and two of them are still growing quickly.

Sanskrit e-text is also being put to use on a scale it never was before. Aggregators such as DharmaNexus, Sañcaya, and the Searchable Aggregate Library of Sanskrit gather copies of many collections and build full-text search on top of them, and in some cases much more, such as grammatical analysis, machine translation, and intertextuality detection. This work does not wait for anyone to map the field first, and it should not have to. But what is known about the field's data has not kept pace with these computational developments. Data scarcity is no longer the problem. The problem is that it is becoming very difficult to say how much data there is, where it is, and how much of it any one project holds.

1.1 Published surveys

Writing shortly before the final additions to GRETIL (October 2020), Meyer (2019) estimated that about 1,500 Sanskrit e-texts were publicly available, some 350 MB of plaintext. He noted that unique texts are hard to count, because several scholars have input the same work, from the same edition or not, and because some texts are subsets of others. More recently, Pradeep and Mamidi (2025) commented on available e-text data for their broad survey of Sanskrit computational linguistics. They concluded that digitized texts and datasets are a primary bottleneck for machine learning, along with the difficulty of comparing results across diverse datasets and metrics. However, their account of digital libraries named only GRETIL, the Sanskrit Library,

SARIT, and Ambuda; the larger collections including E-bharatisampat and Sanskrit Wikisource were not mentioned. Nor did this survey attempt any quantitative description, such as counts of files, texts, or works; sizes; or degree of collection overlap. Meyer's rough count of 2019 remains the only published attempt to count the field as a whole.

1.2 Registers

GRETIL, the Göttingen Register of Electronic Texts in Indian Languages, was started in 2001 by Reinhold Grünendahl as a cumulative register of the many download sites for e-texts in Indian languages; only later did it turn to hosting texts itself (Kulkarni, 2016; Pradeep and Mamidi, 2025, §12.2). In time, GRETIL absorbed or eclipsed other similar register-and-hosting efforts, such as Indology[1] and SanskNet.[2] At the same time, it left a number of substantial collections completely untracked, such as Muktabodha and SARIT. For decades, no register has tried to be for e-texts what the New Catalogus Catalogorum (NCC) tries to be for manuscripts and printed books.

1.3 Aggregators

Over the past few years, especially since GRETIL's intake operations ended in October 2020, several projects have been gathering copies of existing e-texts for specific search and NLP purposes (Table 1).

Primary person	**Project name and data URL**	**NLP purpose(s) and online interface**	**Major collections aggregated**
Harry Spier	*Searchable Aggregate Library of Sanskrit (SALSa)*	Full-text search; `searchable-sanskrit-library.org`	GRETIL, Muktabodha, DCS, SARIT, UTA Dharmaśāstra, Vedic Literature Unicode (Peter Freund)
Vishvas Vasuki	*Vishvas's Website, raw_etexts* `github.com/sanskrit/raw_etexts`	Full-text search; `vishvasa.github.io/full_search`	GRETIL, Muktabodha, SARIT, Peter Freund, **E-bharatisampat** (partial)
Sunder Chakravarty	*Sañcaya*; `github.com/cahcblr/sanchaya`	Full-text search (Zoekt); retrieval for the Patra Darpaṇa MCP server; `sanchaya.rasowshi.us`	GRETIL, DCS, UTA Dharmaśāstra, sanskrit-kosha, Hyderabad DTCS, **Sanskrit Wikisource** (partial)
Giacomo De Luca	*Sanskrit Travelogue, Sanskrit Voyager*; `github.com/SanskritVoyager/SanskritTravelogue`	Retrieval across sandhi and compounds, lemmatization, NLP training data; `sanskritvoyager.com`	GRETIL, Muktabodha, SARIT, DCS, DSBC, DharmaNexus, Yogavaiśāradī, **Sanskrit Documents**
Ryan Conlon	*Intellexus*; `github.com/con-jo-ry/ilx_skt`	Intertextuality detection, conceptual pattern analysis; no online interface	GRETIL, Muktabodha, DSBC, UTA Dharmaśāstra, Hamburg etexts, **Sanskrit Wikisource** (partial)
Sebastian Nehrdich	*DharmaNexus, Dharmamitra*; `github.com/dharmamitra/dharmanexus-sanskrit`	Full-text search, grammatical analysis, machine translation, intertextuality detection; `dharmamitra.org`	GRETIL, Muktabodha, DSBC, and a filtered subset of Intellexus including its **Sanskrit Wikisource**

Table 1: Aggregating projects and the major collections they draw on. Figures were measured in October 2026 against the Sāgarasaṅgama Atlases. Not a comprehensive list.[3]

[1] See `https://web.archive.org/web/20130404022547/http://www.indology.info/etexts/`

[2] Launched around the same time as GRETIL, with 475 books digitized in its first phase (Kulkarni, 2016). The original site no longer resolves. It is the most prominent of the roughly half of GRETIL's external links that are now dead (Neill, 2024), and its files survive only through a private mirror (wilbourhall.org) and in texts that passed into other collections.

[3] For example, a brand new `https://indike.org/pramana` project announced on 19 October 2026 both aggregates collections and produces new OCR of `Archive.org` scans with recent Gemini models (e.g., 3.8 Flash).

These aggregations' common ingredients are generally GRETIL and Muktabodha, as both of them are available as bulk downloads from which it is relatively easy to extract plaintext. The larger collections Sanskrit Wikisource, E-bharatisampat, and Sanskrit Documents are also starting to show up in limited ways.

The case of Sanskrit Wikisource is especially instructive, as its curation problems (see §2.2.1) tend to create obstacles to finding its material. In mid-2026, Intellexus had a student assistant hand-compile a list of Sanskrit Wikisource URLs for export, which yielded 961 distinct texts (1,056 files);[4] most of the material outside Wikisource's properly categorized core is missing. In turn, Dharmamitra took a subset of this same Wikisource material via Intellexus, keeping material with less than 50% character-15-gram overlap with its existing corpus, totaling 685 texts (718 files).[5] Sañcaya drew on Wikisource independently, reaching a smaller number of 165 texts (320 files) but with a volume larger than that of Dharmamitra's import, owing to a few large works such as the Vācaspatyam and Skandapurāṇa.[6]

	All Wikisource 3,749 texts, 818 MB	**Categorized Wikisource 1,669 texts, 459 MB**	**Uncategorized Wikisource 2,080 texts, 359 MB**
Intellexus	961 texts 150 MB · 18%	855 texts 141 MB · 31%	106 texts 9 MB · 2.4%
Dharmamitra	685 texts 89 MB · 11%	608 texts 84 MB · 18%	77 texts 5 MB · 1.5%
Sañcaya	164 texts 101 MB · 12%	140 texts 93 MB · 20%	24 texts 8 MB · 2.2%

Table 2: Sanskrit Wikisource via Intellexus, Dharmamitra, and Sañcaya, by volume. Measured 2026-10-19 against the Wikisource Atlas (dump 2026-10-01). Sizes are the Atlas's own text counts and IAST sizes. Volumes count only what each project actually holds of each text, capped at the Atlas size of that text.[7]

Sanskrit Travelogue, by contrast, left Wikisource out altogether, judging that it "presents just a fraction of the Vedic portion of the Gretil library" (De Luca et al., 2026, §2), although Wikisource has long been broad in scope and is nearly four times GRETIL's size (§2.3). Travelogue did, however, ingest a complete scrape of Sanskrit Documents, greatly inflating its text count in the process.[8] E-bharatisampat is less well known; only Vasuki scraped a partial snapshot[9] of the site as it was in early 2025.

Aggregator projects, then, build selective copies according to both their particular goals and their ability to understand, access, and process potential sources. Nothing guarantees they tell users how much of each source they cover at any given time, or how faithfully, much less how those various sources overlap. Moreover, whether these copies are updated to stay in sync with upstream changes depends on how much processing is required to fit that upstream data to the aggregator project's downstream applications (e.g., full-text search, model training, etc.). In short, aggregators' main strength is the search or NLP system they build on their (usually

[4]Epics were excluded. On measurements of Sanskrit Wikisource Atlas, see §2.2.1. Commit: `https://github.com/con-jo-ry/ilx_skt/commit/f90719f`, 26 August 2026, "wikisource added".

[5]Commit: `https://github.com/dharmamitra/dharmanexus-sanskrit/commit/2181317`, 8 October 2026, "Add 993 texts from the ilx_skt meta-repository (Intellexus)", based on ILX then-latest commit `https://github.com/con-jo-ry/ilx_skt/commit/222bcc4`, 31 August 2026.

[6]320 files, among them the Kathāsaritsāgara as 102 separate taraṅga pages, a work Sañcaya also holds twice more, from GRETIL and from an OCR of the Nirṇayasāgara edition. Nearly all of this is again from the categorized portion; only 24 texts (about 20 MB) come from the uncategorized remainder. These are measured either by clear Wikisource links in their headers or by 50% character-15-gram overlap.

[7]Intellexus took whole texts almost throughout: 905 of 952 measurable texts are ≥80% complete. Exceptions include the Śuklayajurveda (9%) and Padmapurāṇa (59%). Dharmamitra's 50% rule dropped parts of some texts, not just whole texts: of the Skandapurāṇa (15.5 MB IAST in the Atlas) it kept only about 9%. Sañcaya holds several partial texts, e.g. about half of the Vācaspatyam.

[8]"Total unique texts" 12,395, of which Sanskrit Documents contributes 8,614 (69.5%). It's clear that this grand total is in fact not a count of unique texts, as it makes no attempt to deduplicate, e.g., DharmaNexus (1,398) against the GRETIL (800), DSBC (751), and Muktabodha (495) collections from which DharmaNexus is largely built. `https://www.sanskritvoyager.com/docs/sanskrit_travelogue`, accessed 20 October 2026.

[9]In February 2025: 2,696 e-texts, 51% of the 5,126 then online, and 881 MB of 2,103 MB (41%). Relative to today's totals, the current copy holds 2,265 e-texts (41%) and 849 MB (37%).

transformed) data; they are in general an imperfect way to organize primary access to data, especially when that data is being continuously produced in a decentralized manner.

1.4 A missing denominator

Even when a collection or meta-collection can be reliably totaled, assessing the broader significance of this total requires a denominator that the field has never had: the set of all Sanskrit e-texts that exist at a given time. Without it, no corpus can say how quantitatively complete it is. Also, without a comprehensive inventory of what exists, it can be very difficult for an individual reader looking for a searchable e-text to tell whether a work has not yet been digitized or is only hard to find.

To address these needs, this paper presents Sāgarasaṅgama (a "Meeting of the Oceans") and its Atlases,[10] the newest part of the Kalpataru Grove ecosystem of open tools.[11] Section 2 describes how the Atlas-based Sāgarasaṅgama system works and what it reveals. Section 3 sketches how Sāgarasaṅgama will merge with the Sanskrit E-Text Inventory (SETI), which already links the e-texts of smaller scholarly collections to work records in the Pandit database, and so arrive at the missing denominator. Section 4 concludes by looking further ahead, to a second denominator: all the premodern works, as recorded in the NCC, that someday could have one or more e-texts.

2 Sāgarasaṅgama

2.1 A virtual confluence of bespoke finding aids

The foundation of Sāgarasaṅgama is a set of structurally attuned finding aids for large collections, called Atlases. An Atlas collects and reorganizes a collection's metadata into a clearer front end with navigation tailored to that collection. It does not re-host anything, and every link leads back to the collection itself. Three Atlases are published (Sanskrit Wikisource, Sanskrit Documents, and E-bharatisampat) and two more are under active development (Jain eLibrary and GRETIL). Collections that are primarily focused on images as opposed to searchable text are deliberately excluded; for example, eGangotri uploads images to Internet Archive, and Archive's Tesseract OCR is not usable for Sanskrit material.[12] Also excluded are aggregated meta-collections (§1.3 above).

Building an Atlas follows a common pattern. The first step is to ascertain the general shape of the collection, through high-level analysis of its website and, where applicable, its bulk download. The second is to secure the specifics of individual items in order to obtain sizes, dates, and growth patterns. Web scraping is necessary where text cannot be obtained in bulk (Sanskrit Documents, E-bharatisampat, and Jain eLibrary), and it proceeds by automation at a maximum rate of one request every two seconds. All sizes are measured the same way: body text only,[13] with markup and headers stripped, transliterated to IAST, and counted in bytes. Then, the Atlas front end is created with a bespoke browsing structure tailored to the source site, along with added conveniences including transliteration, dynamically responsive search, and graphical insights into growth history. Using the Atlas then deepens the understanding of the source collection, which in turn improves the Atlas, and the result is a detailed write-up and an audit pipeline that tracks the collection's known structural problems (both on the About page).

[10] sagarasangama.info.

[11] kalpataru-grove.info.

[12] AI4Bharat's Sangraha corpus is built largely from Google Cloud Vision OCR of Internet Archive material (Khan et al., 2024).

[13] Exclusion of front material needs to be improved for E-bharatisampat and Jain eLibrary. Both Atlases measure a text by subtraction: they strip markup and recognized page furniture and count what is left, so front matter, publisher boilerplate, and Roman-script apparatus that survive the stripping are counted as text. A planned additional measure would instead count only what is positively recognized as Sanskrit, taking care not to discard romanized texts such as E-bharatisampat's 1.4 MB Romanized Ṛgveda, which contains no Devanāgarī at all.

Finally, a central Sāgarasaṅgama component (`sagarasangama.info`) reads only what each Atlas publishes — a tree of the collection and a dated changelog — presents their aggregated statistics, and offers federated search of titles and, to a lesser extent, authors.

2.2 Atlases

2.2.1 Sanskrit Wikisource

Sanskrit Wikisource (sa.wikisource.org), the Sanskrit branch of the Wikimedia Foundation's library of source texts, has been open to contributors since July 2004. It is the largest single collection whose text can be bulk-downloaded under an open license (CC BY-SA 4.0). It overtook GRETIL in size in the first half of 2017, after a burst of growth that began in 2016. One strength of the system is near-perfect versioning: because Wikimedia keeps every revision of nearly every page,[14] the collection can be reconstructed as it stood on any day. A weakness of Wikisource for corpus building is that it has no notion of a "text," only pages which link freely to one another in whatever way a given contributor decides, including by way of categories. Similarly, although categorization is the backbone structure for browsing (see *vargasarvasvam* page), all categorization is done manually and subject to error, and much material, including most unproofread OCR, remains outside the main category tree. Text from the built-in OCR pipeline can be fed directly into user-facing ("Mainspace") pages through an import mechanism called "transclusion."

The Atlas uses this versioning to reconstruct the collection month by month back to 2012, on the same definitions it uses today.[15] To create a notion of "text," the Atlas interprets various subpage structures — slash-separated titles (from which Wikisource generates so-called "breadcrumb" links), transclusions, and so on — rolling them and their associated statistics up into single items, with substructure retained as notes (Figure 1).[16] To highlight material sitting outside of the ideal category tree — where each page is categorized and each category itself has exactly one category parent — the Atlas creates an artificial category called "improperly categorized" (*asambaddhavargīkṛta*) for orphaned categories and uncategorized pages, which can optionally be left out of totals. In addition, an audit lists problems that should be fixed on the site itself, including 60 orphaned categories and 699 pages that look like parts of a text but lack a working slash title (e.g., "samarāṅgaṇasūtradhāra adhyāya 1").

2.2.2 Sanskrit Documents

Sanskrit Documents (sanskritdocuments.org) has been online since the 1990s, which makes it one of the oldest Sanskrit e-text collections on the web. Its 9,756 documents total only 132.6 MB, and the median document size is a mere 2.8 KB. This is because most items are short devotional pieces such as stotras and nāmāvalīs, and just fourteen texts (the Purāṇas and Monier-Williams among them) carry a sixth of the total. Its organization is unusually layered, with three cross-cutting categorizations. Each document lives in exactly one of twenty server folders. Eighty-five topic pages, which the site's menus link to, draw documents from across those folders, so that a document appears on about two topic pages on average. And each document carries a free-text category line, with 317 distinct values after spelling normalization; shivarahasya alone tags 784

[14] Deleted pages are an exception. The Atlas rebuilds each past month from Wikimedia's full revision-history dump, and a page deleted before that dump was taken is missing from it altogether, so each reconstructed month is a slight undercount. The loss appears small: of 5,199 removals in an earlier series built from archived monthly dumps, only about 106 (2%) were genuine deletions, the rest being renames and re-filings. The Internet Archive's 76 monthly dumps (September 2011 to May 2022) could recover many deleted pages if needed. Pages also appear under their current titles, so a page renamed in 2020 appears renamed in 2013 as well.

[15] Sanskrit Wikisource opened on 23 July 2004, but the Atlas's tree descends from the root category *vargasarvasvam*, which was created only on 20 January 2012. Before then the site had at most two categories, so there is no tree to rebuild. Counted without categories, the collection grew from 9 pages on its first day to about 1,900 pages and 10 MB of IAST by July 2011.

[16] Besides slash titles, the Atlas follows redirects, tolerates stray whitespace, and nests the hyphenated chapter titles of the Mahābhārata and the Ṛgveda by explicit rule, but it does not guess at other separators. Pages it cannot attach (see the audit below) currently count as separate texts, so the number of texts is overcounted; sizes are unaffected.

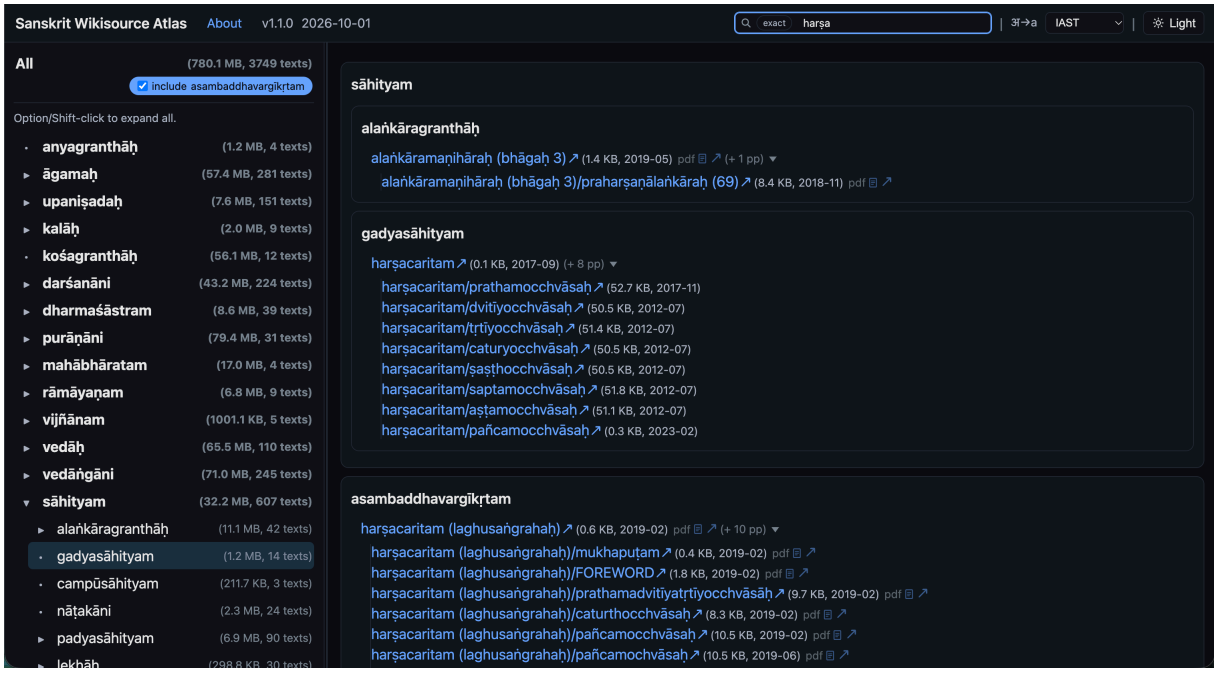


Figure 1: Sanskrit Wikisource Atlas (v1.1.0, data of 1 October 2026), search for "harṣa". Left: the category tree, with sizes and text counts for each category and the improperly categorized branch (*asambaddhavargīkṛtam*) included in the overall totals. Right: the Harṣacarita's ucchvāsa subpages rolled up into a single text under *gadyasāhityam*, and a separate abridgment of it (*laghusaṅgrahaḥ*) that sits outside the main tree.

documents, which is significantly more than the smallest topics. Dating is weak: the only date a document records is its last update, so edits pull documents toward the present.

The Atlas offers folders and topics as alternative trees and turns the category line into a filter. Two earlier copies of the site, a June 2020 GitHub snapshot and my own February 2025 scrape, partially correct the weak dating by proving that 153 documents were already present earlier than their stamps say. The Atlas also gathers the scan links the site gives, which reduce to 873 distinct sources, though the site never claims that these were the exemplars used for producing the e-texts.

2.2.3 E-bharatisampat

E-bharatisampat (ebharatisampat.in), the digital library of Samskrita Bharati, is the largest of these collections by far and the least known. Of its 11,066 items, 5,529 carry searchable text totaling 2,321 MB, more than Wikisource, GRETIL, and Sanskrit Documents combined. It has also been the fastest-growing in recent years: the earliest items the Atlas can date were added in January 2021, and by the end of that year there were about 2,000 items, against more than 11,000 by August 2026. Searchable text has not kept pace, however: since 2024 roughly half of new items have been PDFs without text, and in 2026 almost all of them.[17] Where text exists, it often appears to be hand-corrected OCR, including large amounts of front material. Categorization is a simple hierarchy, with every item in exactly one subcategory under one category, although placement and author names are not consistent (e.g., versions of the Kādambarī are spread across prose, verse, and story). Reading — with gradual but automatic pagination — is publicly open, but one-click text downloads generally require a login, and the site's "Terms & Condition" [sic] link is broken. The site publishes no dates.

[17] A recheck in October 2026 of every item uploaded in 2026 as a PDF without text found that 9 of 2,494 had since gained searchable text, all of them uploaded between January and early March 2026. Text is thus added after upload, apparently in retrospective batches, but so far only slowly.

The Atlas infers dates from two implicit signals that agree closely (Spearman $\rho = 0.972$): serial numbers and the timestamps hidden in thumbnail filenames. More significantly, the Atlas makes a substantial amount of hidden material visible. About a third of the items that have both text and a PDF (1,365 of 3,912) never appear when browsing the site's text section, and subcategory browsing undercounts further. For example, Āyurveda shows 32 items where the Atlas finds 187. Measuring size required several weeks of scraping at one request every two seconds, since many texts are paginated into hundreds of separate loads.

2.2.4 Jain eLibrary

Jain eLibrary, run by the education arm of JAINA, is a large and mostly Gujarati and Hindi library of scanned books, with some 41,600 catalog items. Its PDFs can be downloaded only with an account, but the library is otherwise liberal about sharing its data. Its associated search engine, Jain Quantum (jainqq.org), created by Anish Visaria, publishes the raw OCR text of each item it indexes on a single public page, with no login required. This OCR text includes large amounts of front material, etc. Quantum's coverage ends in June 2020, but the eLibrary has its own, competing set of raw OCR files, made in a single batch in July 2022 and available only with an account. For items added up to June 2020, which have both, neither version is clearly better. Books added after 2022 have no text from either source. The catalog's classification leads with codes for format and script (e.g., "Book_Devnagari", "Book_English") and describes content only through loose keywords, and its author field often names the editor or translator, with the original author in a separate field that is often empty.

The Atlas primarily covers books cataloged as Sanskrit alone,[18] and secondarily those cataloged as Prakrit. OCR content is taken from Quantum where available, otherwise from the eLibrary. The Atlas allows readers to browse either by author, separating the classical author from the editor a book was filed under where the catalog allows, or by the Śvetāmbara Āgama canon, using the library's own canon tags (on 455 books). Dates are measured from the day each item went online, although their text generally came online later.

2.2.5 GRETIL

GRETIL, the Göttingen Register of Electronic Texts in Indian Languages, was published by the Göttingen State and University Library from 2001 to 2020, declared closed in July 2025, and archived to TextGrid. It is famously open, with its core TEI files under a Creative Commons license. These TEI XML files were for the most part converted from legacy files between 2017 and 2020, and they are stored flat, with no category structure. In addition, GRETIL also exists as an older tree of legacy HTML files, sorted into category folders on the web server and in the cumulative download served by the website. Neither of these sets contains the other: 30 texts exist only as legacy HTML (e.g., Mahābhārata, Śatapatha Brāhmaṇa), while 19 were born as TEI. GRETIL's update history page logs its updates from November 2001 to September 2020. Texts that were already online when the log began, including the Mahābhārata and much of the Veda, never appear there as additions, so the Atlas cannot date them.

The Atlas is built from a November 2025 scrape of the live site, cross-checked against the downloadable zip and my own mirror, and its links lead to that mirror. For its text count, the Atlas associates every legacy file, including alternative renderings (e.g., BHELA sandhi and compound analysis) and indexes, with the text it belongs to, and counts those texts rather than files; alternative renderings and indexes are not counted toward cumulative text or size totals.

2.3 What the Atlases reveal

Size. Together the five collections hold 22,129 texts and 4.9 GB of IAST plaintext (Table 3). Even leaving aside Jain eLibrary, whose text is uncorrected OCR, and Sanskrit Documents,

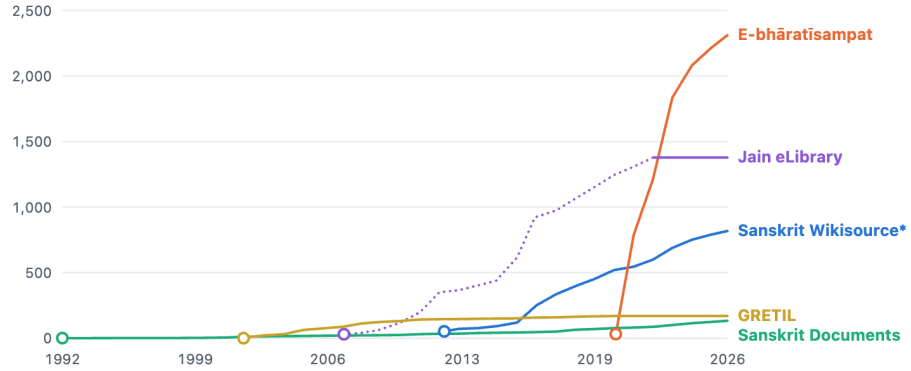
[18] Books cataloged under mixed language strings are not yet included, although they are the larger group. Among Sanskrit-tagged books in the library's catalog data (October 2026), 1,750 are cataloged as Sanskrit alone, against 2,383 as "Sanskrit, Hindi", 892 as "Hindi, Sanskrit", and 842 as Sanskrit with Gujarati.

whose texts average only about 14 KB (against 200–420 KB for the others), the remaining three collections hold 10,090 texts in 3.4 GB.[19]

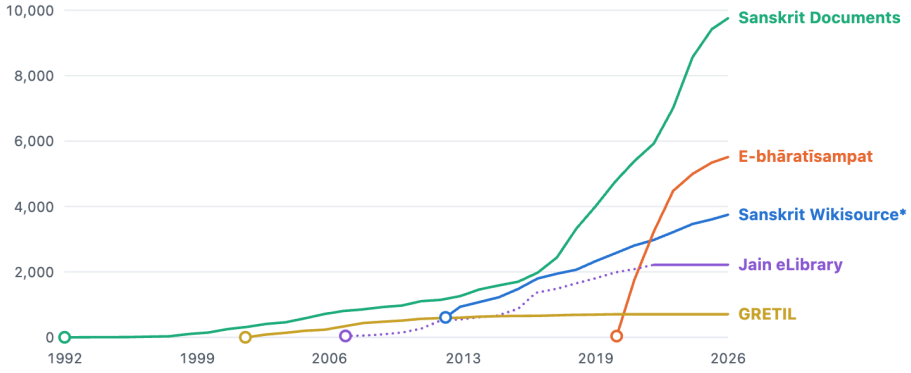
Collection	Texts	Size (MB)	Online since	Data as of
E-bharatisampat	5,529	2,321	2021 (earliest datable item)	Sept 2026
Jain eLibrary	2,285	1,419	2007	Oct 2026
Sanskrit Wikisource	3,749	818	2004	Oct 2026
GRETIL	812	218	2001 (intake ended 2020)	Nov 2025 (scrape)
Sanskrit Documents	9,754	133	1990s	Aug 2026
Total	**22,129**	**4,908**		

Table 3: The five collections as measured by their Atlases. Sizes are IAST bytes of body text.

Growth. Because each Atlas dates its items, Sāgarasaṅgama can show how the collections grew relative to each other (Figure 2).



(a)



(b)

Figure 2: Growth of the five collections, 1992–2026, from the Atlases' dated records (Sāgarasaṅgama, "Growth over time"): (a) cumulative size of IAST plaintext in MB; (b) cumulative number of texts. Circles mark each collection's earliest dated item. Jain eLibrary's line is dotted up to the library's own OCR batch of July 2022, counting books that were online before that text was made. *Sanskrit Wikisource's text counts for parts of 2014 and 2015 are corrected for a double count in the upstream snapshots.

[19] E-bharatisampat's sizes will need to be adjusted downward once front material and similar furniture are more properly removed (see §2.1). The same holds for Jain eLibrary, whose OCR also keeps its noise and, in many books cataloged as Sanskrit alone, a Hindi apparatus in Devanāgarī.

The comparison with earlier estimates is striking. On 1 January 2019, the year of Meyer's count of about 1,500 texts and 350 MB, Sanskrit Wikisource alone held 2,058 texts and 402 MB. Wikisource, Sanskrit Documents, and the dated part of GRETIL together held 6,138 texts and 630 MB;[20] Jain eLibrary reached 1,670 books, but its OCR was yet to come. E-bharatisampat, now the largest of the five by far, did not yet exist. By 2022, however, both had their text online. In sum, the estimates have trailed the collections for years, and the gap widens by the month.

Structure. Each collection has a unique structure, and none presents its structure satisfactorily. Sanskrit Wikisource has pages and hand-assigned categories but no notion of a text; the Atlas supplies one, and it turns the uncategorized remainder into an explicit branch rather than an absence. Sanskrit Documents has three cross-cutting organizations (server folders, topic pages, and a free-text category line), none of them primary; the Atlas lets the reader clearly switch between the first two and filter by the third. E-bharatisampat has a single, simple hierarchy, but its browsing pages do not show everything filed in it; the Atlas remedies this by showing every item where it was filed. Jain eLibrary's catalog classifies books by format and script, describing content only through loose keywords; the Atlas regroups the books by author and by canon. GRETIL survives as two overlapping layers, a flat set of TEI files and an older tree of legacy HTML; the Atlas reunites them as one list of texts, complete with the category structure, without reintroducing the noise of ancillary files. In each case the result is not a new classification imposed from outside, but the collection's own relevant structure, made complete and navigable.

Hidden material. As a special case of structural issues, a given collection interface can sometimes make certain material so difficult to find that it is virtually hidden. In Sanskrit Wikisource, more than half of all texts (about 2,000 of 3,749) are uncategorized and so unreachable from browsing the main category tree; besides this, hundreds of pages stand apart from the texts they belong to. In Sanskrit Documents, a small number of texts are served by the site (e.g., included in search results) but not listed by topic or location pages. Finally, E-bharatisampat simply fails to display well over a thousand e-texts in the site's text browsing section, and on top of that, subcategory browsing undercounts badly. These properties of the source collections pose substantial obstacles to individuals and aggregators alike: a project that collects what it can see will very likely miss what the source itself does not readily show.

2.4 sagarasangama.info

The site sagarasangama.info, launched in September 2026, is the interface over all five Atlases. Since five collections organized in five incompatible ways share no hierarchy that could be browsed together, its main interface is search: it searches titles (and to a lesser extent authors) across all five collections at once (Figure 3), allows grouping by Atlas, and offers fuzzy search (especially helpful for Jain eLibrary, where essential metadata fields like title use non-standard Roman transliteration). Each result keeps what its own Atlas gives it, such as size and date statistics and links to associated PDFs. Finally, every query and filter is kept in the URL, so that a list of results can be shared or cited. Because the data are versioned publicly in Git, a result seen on a given date can be traced to a specific commit.

On its About page, the site also shows the combined counts and growth series of the Atlases, which are refreshed whenever an Atlas publishes new data.

[20] These are the last Atlas data points on or before 1 January 2019. GRETIL's figure excludes its 109 undated works, and because Sanskrit Documents records only last-update dates, its figure is a lower bound.

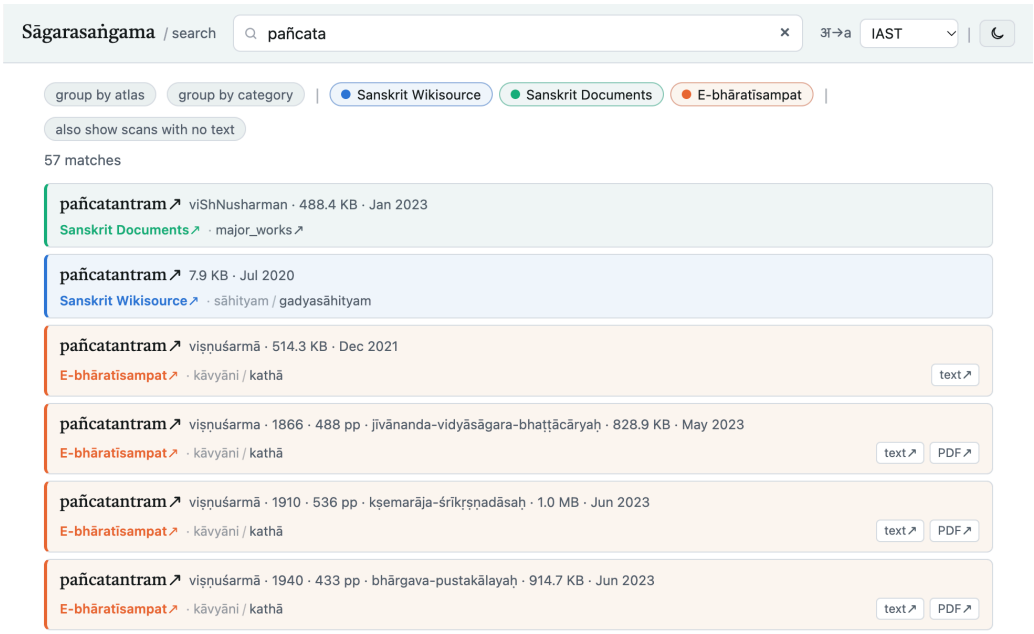


Figure 3: Sāgarasaṅgama search for "pañcata" (57 matches), here returning results from Sanskrit Documents, Sanskrit Wikisource, and E-bharatisampat.

3 Toward one inventory of all extant e-texts

The denominator this paper is after, the set of all Sanskrit e-texts that exist at a given time — call it D1 — is a function of three things: 1) ongoing attention to (typically large) collections that require intensive analytical treatment, 2) ongoing attention to (typically small) collections that do not, and 3) a way of identifying and subtracting text duplicates within and across collections. The first is the purview of Sāgarasaṅgama Atlases (§2.2), which can increase in number as necessary. The second is the immediate purview of SETI (§3.1), which itself must expand in two ways: 1) clean and otherwise improve coverage data for collections it has already reached, and 2) cover unreached collections for the first time. Once an Atlas has illuminated a complex collection, the associated metadata is then ready for ingestion into SETI (§3.2). The third and final major step toward D1, deduplication, mainly requires extending SETI with further annotations, including which edition a given e-text is based on and how good the e-text is. With that in place, different end users can develop and apply different similarity metrics and exclusion thresholds for their specific purposes.

3.1 SETI

SETI, the Sanskrit E-Text Inventory (Neill, 2025), revives GRETIL's original function as a pure register, recording where searchable Sanskrit e-texts are available online without hosting any of them. Its foundation is an open spreadsheet, exposed through an API, that enumerates e-texts by collection and maps them to locations online. SETI also aligns each e-text with one or more work identifiers from the Pandit prosopographical database, which function as authority records. SETI now lists 1,747 e-texts in eight scholarly collections (including GRETIL),[21] of which 1,271 (73%) are matched to Pandit work IDs.

[21] Slated for addition are the rest of Muktabodha's collection, the Todai Bussei Electronic Text Library, the Digital Sanskrit Buddhist Canon, Sanskrit Documents, Wisdom Library, Ambuda, and Sanskrit Sahitya; the all-Prakrit PDTP awaits a wider expansion into those associated languages.

3.2 Merging Sāgarasaṅgama into SETI

Each Atlas's enumeration of its collection's texts will be added as new rows to SETI. This action alone will make SETI the most complete register of Sanskrit e-texts online.[22]

Other fields have built registers of this kind. The Perseus Catalog (Babeu, 2019) and the Digital Latin Library Catalog (Huskey, 2019) register online editions of Greek and Latin works, and the resources for Kanjur and Tanjur studies (rKTs) register the witnesses of more than a hundred Kanjur and Tanjur collections, locating each by volume and page in the Buddhist Digital Resource Center's (BDRC) scans and showing associated e-texts beside them (Griffiths, 2025).[23] The DLL Catalog's fate shows what such a register needs most: upkeep that continues as long as the collections keep growing.[24]

Enumeration is only the first step. What makes a given register item more or less useful to a given user is chiefly a combination of the edition it is based on and its quality. Some collections do not volunteer information about the edition exemplar, and for some e-texts there may not be a fact of the matter if they mix multiple sources with or without saying so. Copies in the strict sense — the same transcription of the same edition carried from one collection to another — certainly occur (E-bharatisampat, for instance, holds GRETIL files simply converted to Devanāgarī), but their extent is not yet reliably known; clear annotation of borrowing is not guaranteed, and borrowings may also be modified, so automatic detection is not always straightforward either. The choice of what to identify as a duplicate also depends on purpose: one user will want to choose the better edition at lower transcription quality, another the worse edition at higher quality, and another might want to keep both provided they are sufficiently distinct. The inventory should therefore record edition and quality and let its users choose, rather than discard anything itself. In turn, teams can choose the deduplication metric and threshold that works for their purpose (e.g., Dharmamitra's recent use of character-15-gram overlap and a 50% cutoff).

For recording editions there are also models. BDRC's catalog, following the BIBFRAME model, attaches each e-text and scan to a particular version (a print, blockprint, or manuscript) of a work (BDRC, nd), and CTS identifiers distinguish editions of a work as well as the work itself (Blackwell and Smith, 2019).

Note finally that the inventory is also dated: the Atlases date their items, reaching back years before SETI existed, and the SETI master sheet is tracked in Git from here on. D1 is therefore not one number in two senses: it is a series over time, and at any given time there are as many D1s as there are ways of deduplicating. Any corpus, aggregated or not, can then state its coverage against the D1 that fits its purpose, say which one that is, and state it again a year later.

4 Conclusion: from e-text accounting (D1) to work representation (D2)

Sanskrit e-text is now far larger than the field's published picture of it, and the projects that build on it hold partial copies whose coverage no one has measured. The Sāgarasaṅgama Atlases give the first measured view of five very large collections: how large they are, how they grew, and what their own interfaces hide. Extending SETI to ever more collections, including Sāgarasaṅgama's current and future Atlases, will yield D1, the set of all extant e-texts, dated and annotated by edition and quality. What the field most needs standardized is not its data but access to it, and an inventory of this kind standardizes access without asking any collection to change.

[22] In the other direction, SETI's non-Atlas collections with less complex structures will receive a single combined pseudo-Atlas, so that Sāgarasaṅgama can sum over them as it already does over the Atlas-bearing ones.

[23] For the Derge, Lhasa, and Lithang Kanjurs, rKTs links to e-texts made by ACIP, ADARSHA, and Esukhia.

[24] Only 577 of DLL's works (11%) link to scans at all, far fewer than the Perseus Catalog's 1,247 Latin works with scan links in 2013. So the 2019 DLL apparently hadn't absorbed the Perseus Catalog's links in 2019. Its e-text links also come from just four sites: PHI 832, digilibLT 457, Perseus 330, Biblioteca Italiana 145. Corpus Corporum and the LLT — Brepols' Library of Latin Texts, the Latin near-equivalent of the Thesaurus Linguae Graecae or TLG — aren't among them. DLL's development was later frozen, and in April 2026 the catalog announced its transition to a static archive, with "tens of thousands" of item records never processed.

D1 helps people find individual items as well as answer how complete a given corpus is. It does not answer how much of premodern Sanskrit literature is online. For that, a second denominator is needed: all the premodern works that could have an e-text (D2). Here the unit is the work, and Pandit's work identifiers, already attached to most of SETI's e-texts, are the natural keys. However, the most complete authority list of works is not Pandit — which grew outward from Potter's *Bibliography of Indian Philosophies* (Potter, 1970) — but rather the New Catalogus Catalogorum (NCC), which attempts to record every work attested in manuscript or print. Subsequent right-merges of SETI onto Pandit and then onto the NCC would associate NCC entries with Pandit's unique identifiers and SETI's attested e-texts.[25] In this way, we would arrive at the share of known premodern works that are online in any form. Pāṇḍitya already shows how useful this can be: it lets users explore Pandit's works visually, with SETI's e-texts overlaid, and so look up online witnesses work by work (panditya.info; Neill, 2025; Figure 4). The works that such a join finds without any e-text then become an attractive worklist for anyone digitizing Sanskrit, around which a collaborative discussion can develop. Especially for the limited case of printed editions — whose digitization is a far more tractable problem than that of manuscripts, and which form the basis of modern scholarship — complete digitization of that literature becomes an attainable goal, provided we develop ways to coordinate our decentralized efforts for maximum impact.

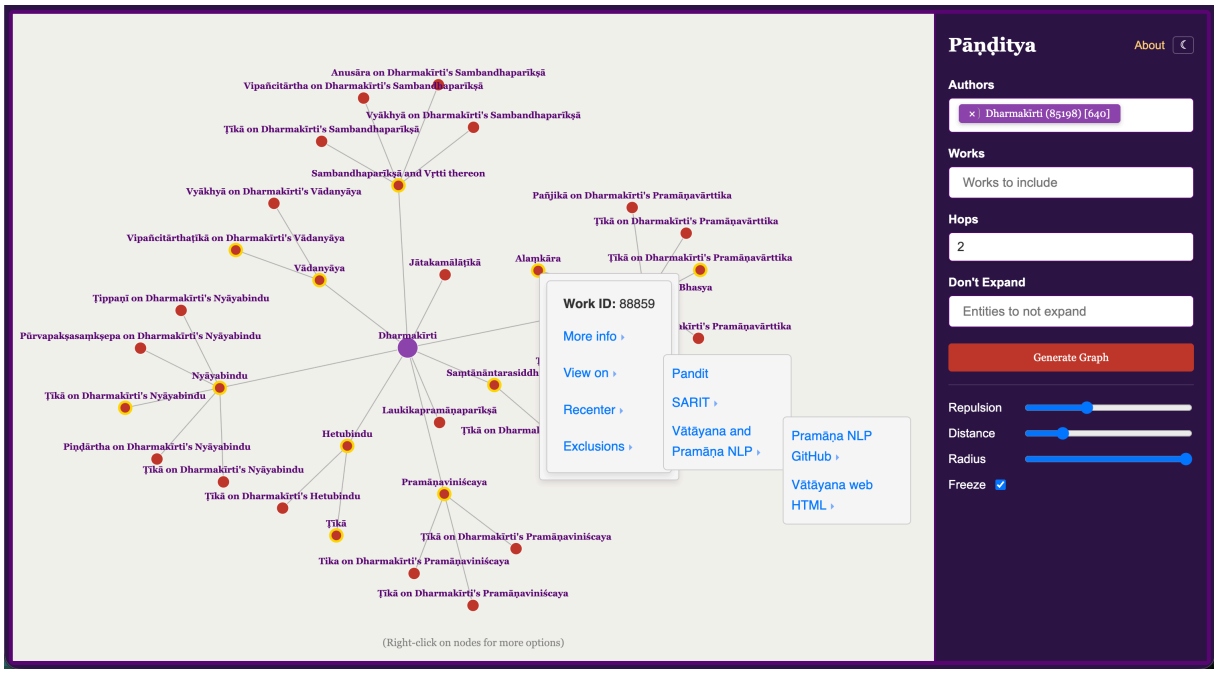


Figure 4: Pāṇḍitya graph of works by Dharmakīrti and commentaries thereon (Pandit author 85198, two hops). Yellow-ringed nodes are works with at least one e-text registered in SETI. The right-click menu for Prajñākara's Alaṃkāra(bhāṣya) (Pandit 88859) links to Pandit and to the available e-texts.

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[25] The obstacle is not the join itself but its keys: Pandit already cites the NCC for about 29% of its works (3,933 of 13,800), but only by volume and page. The join will need either NCC entry identifiers in Pandit, or a digitized NCC headword index matched to Pandit by name, the way SETI was matched to Pandit.

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